

Sharan Teja Medikar

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SUMMARY

Recently completed an MSc in Artificial Intelligence at the University of Surrey, with hands-on experience building AI/ML systems using Python, PyTorch and LLMs. Experienced across applied AI, machine learning, LLM evaluation, computer vision and software engineering, with a particular interest in building dependable AI systems for real-world problems. Seeking AI Engineering and Applied AI opportunities combining technical depth with end-to-end ownership. Currently have the right to work in the UK.

EDUCATION

MSc in Artificial Intelligence | University of Surrey | Sep 2025 - Sep 2026 | Result pending

- **Modules:** Fundamentals of Machine Learning, AI and Sustainability, Ethics and Regulations of AI
- **Dissertation:** Self-Refining Code Generation Using Execution Feedback

BTech in Computer Science and Engineering | National Institute of Technology (NIT), Silchar | Jun 2021 - May 2025

- **Modules:** Machine Learning, Data Structures and Algorithms

WORK EXPERIENCE

Founder & Technical Lead — Police Night Patrolling System | Government of Assam, Shri Bhumi District | Assam, India | Dec 2023 - Sep 2024

- **Conceived and proposed** a real-time police patrolling platform to district government stakeholders, secured approval and assembled a five-person engineering team to deliver it from idea to operational handover.
- **Led development** of real-time tracking supporting 30 patrol vehicles and 60+ operational users, contributing to authentication, backend and real-time data functionality using Flutter, Firebase and Express.js.
- **Led testing, UAT and technical handover** for district-level operational deployment.

Software Engineering Intern | Fleckor Tech | Hyderabad, India | Jun 2024 - Jul 2024

- **Developed frontend and backend components** contributing to 3+ core features including API integration and application functionality under sprint timelines.
- **Identified and resolved approximately 15-20 functional bugs** and supported deployment preparation and application testing.

PROJECTS

Self-Refining Code Generation Using Execution Feedback | Python, PyTorch, Qwen2.5-Coder, CodeLlama

- **Built an execution-feedback code-generation system** that generated solutions, executed tests and iteratively refined failures using structured runtime feedback.
- **Improved Qwen MBPP pass@1 from 71.9% to 82.44%** using adaptive execution-feedback refinement while retaining executable evaluation and bounded stopping.
- **Evaluated single-pass, best-of-5, adaptive and fixed-iteration refinement** across Qwen2.5-Coder and CodeLlama on HumanEval and MBPP, measuring reliability through executable outcomes.
- **Analysed convergence and failure modes** to compare when additional execution-feedback iterations improved or failed to improve generated solutions.

AI Job Application Agent | Python, Playwright, SQLite, Agentic Workflows

- **Built an agentic job-search and application system** combining vacancy discovery, deterministic filtering, semantic matching, evidence-grounded preparation and browser-based ATS interaction.
- **Implemented persistent workflow state, duplicate prevention and validation gates** with human-in-the-loop controls for ambiguous or security-sensitive actions backed by automated tests.
- **Separated deterministic safety logic from semantic judgement** while preserving source provenance and auditable state transitions across the workflow.

Vision-Language Model Prompt Learning and Adaptation | CLIP, CoOp, CoCoOp, LoRA, Prompt Tuning

- **Implemented and evaluated CLIP adaptation and prompt-learning methods** including CoOp, CoCoOp, KgCoOp, PromptSRC, ProGrad, LoRA, Tip-Adapter, CLIP-Adapter, MaPLe and fusion across EuroSAT and Flowers.
- **Improved EuroSAT from 53.99% zero-shot accuracy to 89.70% with Attention Fusion** while CoOp achieved 91.23% Top-1 accuracy on Flowers across remote-sensing and fine-grained visual domains.
- **Compared adaptation baselines on EuroSAT** with LoRA achieving 64.84%, CoCoOp 83.14%, ProGrad 84.99%, KgCoOp 85.59% and Fusion MLP 88.98% accuracy.

Physics-Informed Environmental ML | Python, PyTorch, XGBoost, LightGBM

- **Developed a physics-informed LSTM over 28,632 site-year observations across 26,446 locations** achieving RMSE 32.053 and R^2 0.291 on temporal 2024 evaluation.
- **Improved RMSE by 34.5% over XGBoost and 9.7% over a standard LSTM** while evaluating compression, interpretability and environmental-equity behaviour.
- **Reduced model size with float16 quantisation** from approximately 1.8 MB to 908 KB with essentially no RMSE degradation.

SKILLS

Programming & Data: Python, SQL, JavaScript, NumPy, Pandas

AI & Machine Learning: PyTorch, scikit-learn, Computer Vision, LLM Evaluation, CLIP

Generative AI: Qwen, CodeLlama, AI Agents, Prompt Engineering

Engineering & Tools: APIs, Testing, Git, Linux, Firebase